ENGINEERING PROJECT REPORT · DESIGN & ANALYSIS OF ALGORITHMS

RescueGrid: Disaster Relief Optimization System

Algorithmic Optimization Framework for Multi-Incident Response & Emergency Transport Logistics

COURSE SUBJECT	IMPLEMENTATION STACK	GRAPH TOPOLOGY	VERIFICATION STATUS
Design & Analysis of Algorithms (DAA)	React 18 · TypeScript · Vite · SVG	13 Municipal Nodes · 19 Road Edges	28 Unit/Acceptance Tests Passed

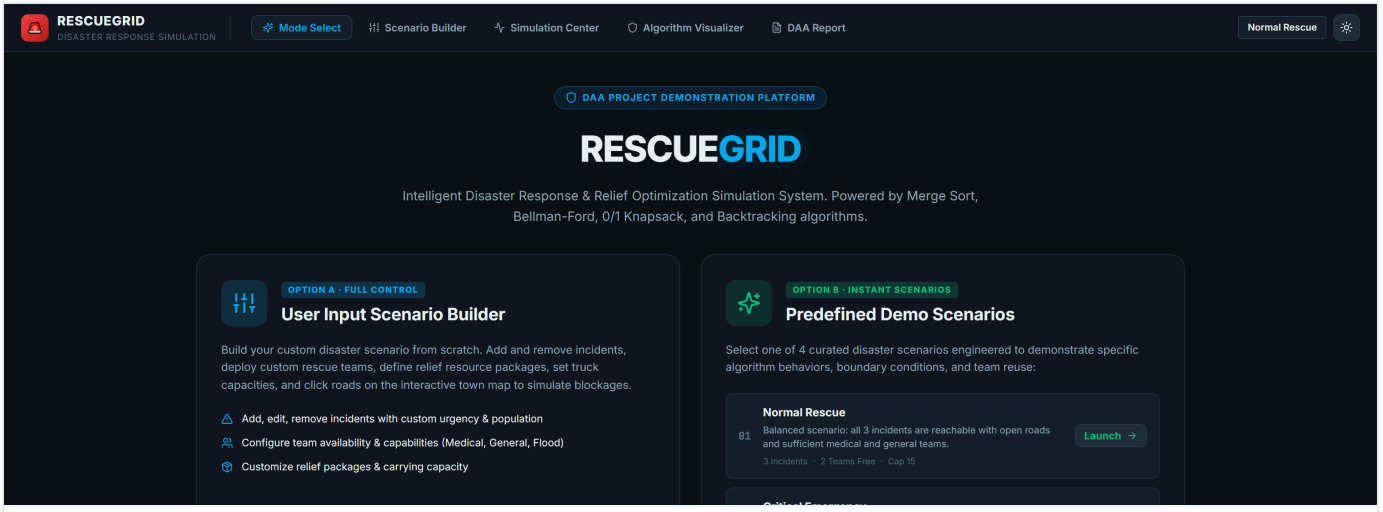
1. Project Overview

RescueGrid is an intelligent disaster relief optimization and simulation system designed to solve high-stakes resource allocation and transit routing dilemmas during acute humanitarian crises. In real-world catastrophes—including torrential river floods, structural collapses, and industrial hazards—emergency management teams confront three fundamental bottlenecks: *triaging casualty incidents* under severe time urgency, *finding safe transit paths* across fractured road networks with collapsed bridges and road blockages, and *packing rescue transports* to maximize humanitarian impact without exceeding strict vehicle payload capacities.

RescueGrid addresses these challenges through a unified discrete-event simulation architecture combining four core DAA algorithmic paradigms: **Merge Sort** with multi-criterion priority heuristics for incident triage, **Bellman-Ford** for resilient shortest-path transit under physical road blockages, **0/1 Knapsack Dynamic Programming** for optimal relief supply cargo packing, and **Backtracking with State Space Search** for constraint-satisfaction rescue team assignments.

Dual-Mode Operational Architecture: The application supports two distinct operational modes: **Option A (User Input Scenario Builder)** presents a completely clean canvas (0 pre-filled demo incidents, 0 teams, 0 packages) allowing operators to construct custom emergency scenarios from scratch, add specialized units, and click roads directly on the interactive town map to simulate blockages. **Option B (Predefined Demo Scenarios)** provides 4 curated emergency cases (Normal Rescue, Critical Emergency, Resource Shortage, and No Route Available) engineered to benchmark boundary conditions, bridge collapses, and supply deficits.

Figure 1: Platform Mode Selection (Option A Custom Builder vs Option B Predefined Emergency Cases) LANDING PORTAL



Functionality: Serves as the landing control center allowing operators to choose between Option A (full control user-input scenario builder) and Option B (instant access to 4 curated emergency response scenarios).

2. Four Algorithms Implemented in RescueGrid

Every algorithm implemented in RescueGrid has been verified directly from the project source code (`src/algorithms.ts`). Each algorithm plays a specialized, mathematically rigorous role in the emergency management pipeline:

1. Merge Sort (Multi-Criterion Triage) `O(R log R)`

How it works: A stable divide-and-conquer sorting algorithm. Recursively halves the incident array until singletons remain, then merges sublists in prioritized sequence using a custom 5-tier comparator function.

Comparator Hierarchy: (1) Urgency Tier (Critical > High > Medium > Low), (2) Medical Emergency flag (true > false), (3) Affected Population (descending), (4) Elapsed Waiting Time (descending), and (5) Deterministic Incident ID.

Role in RescueGrid: Sorts the operational triage queue so that life-threatening medical emergencies in high-density areas receive immediate response first.

Complexity: `O(R log R)` time across all cases; `O(R)` auxiliary space (where R = number of incidents).

2. Bellman-Ford (Shortest Safe Route) `O(V · E)`

How it works: Dynamic programming graph traversal. Initializes distance estimates from HQ to all nodes as ∞ (dist[HQ]=0). Performs |V|-1 passes over all open directed edges, relaxing edges where `dist[u] + cost(u,v) < dist[v]`. Detects unreachable nodes and verifies absence of negative cycles.

Role in RescueGrid: Filters out blocked roads (hazard zones/collapsed bridges) and discovers the guaranteed lowest-cost safe travel route from Operations HQ to any incident site.

Complexity: `O(V · E)` time (V = 13 nodes, E = 38 directed road edges); `O(V)` space for distance and parent vectors.

3. 0/1 Knapsack (Relief Allocation) `O(N · W)`

How it works: Dynamic programming optimization. Builds an (N+1)×(W+1) table where `dp[i][w] = max(dp[i-1][w], value[i] + dp[i-1][w-weight[i]])`. Backtracks from dp[N][W] to recover chosen packages. Benchmarks against a greedy heuristic.

Role in RescueGrid: Emergency vehicles have rigid payload limits (default 15 capacity units). Knapsack packs the most impactful mixture of Trauma Kits, Mobile Clinics, Food Rations, and Water Supplies to maximize casualty support.

Complexity: `O(N · W)` pseudo-polynomial time and space (N = package items, W = payload capacity limit).

4. Backtracking (Team Assignment) `O(M^N) Pruned`

How it works: Depth-First Search exploration of the combinatorial state space tree. At each decision node, candidate teams are evaluated against constraints: team availability and matching capability (medical, flood, structural, general).

Bounding & Pruning: Infeasible branches are pruned immediately (dead end) and the algorithm backtracks up the tree, avoiding unproductive subtrees.

Role in RescueGrid: Assigns available rescue teams to triage incidents without double-booking while ensuring specialized medical/flood skills match the incident needs.

Complexity: `O(M^N)` worst-case; heavily pruned in practice by capability and availability bounding constraints.

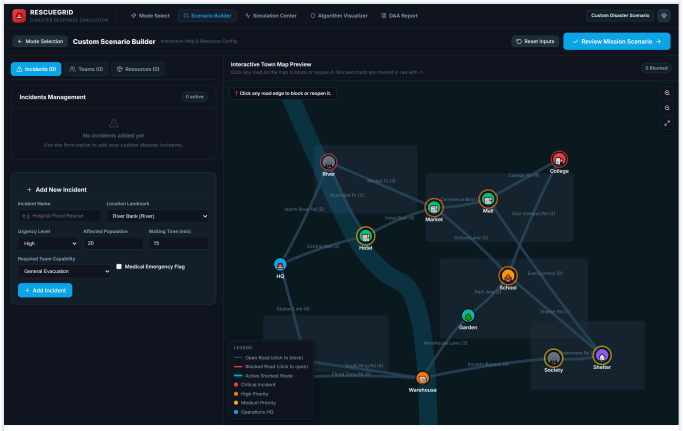
Summary Comparison of Implemented DAA Algorithms

Algorithm Name	Design Paradigm	Time Complexity	Space Complexity	System Responsibility in RescueGrid
Merge Sort	Divide & Conquer	`O(R log R)`	`O(R)`	Multi-criterion priority ordering and triage hierarchy of casualty incidents
Bellman-Ford	Dynamic Programming	`O(V · E)`	`O(V)`	Shortest safe path calculation from Operations HQ bypassing blocked road hazards
0/1 Knapsack	Dynamic Programming	`O(N · W)`	`O(N · W)`	Optimal cargo manifest selection maximizing relief points under vehicle capacity
Backtracking	State Space Tree Search	`O(M^N)` (pruned)	`O(N)` stack	Constraint-satisfaction assignment of specialized teams (Medical, Flood, General)

3. Application Screenshots & Screen Analyses

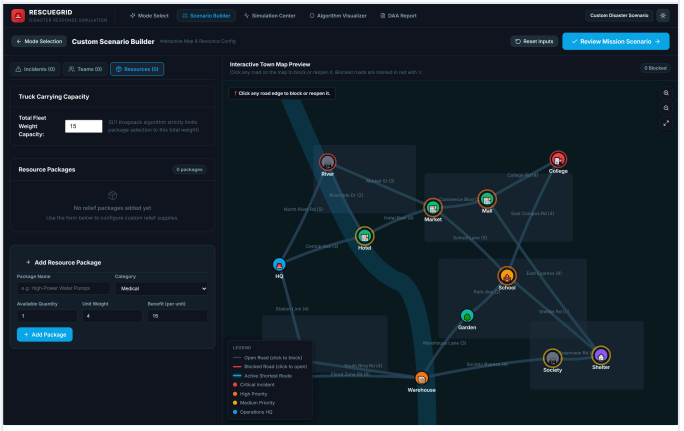
Every visual artifact presented below has been captured directly from the live running application without modification:

Figure 2: Clean Scenario Builder Canvas — OPTION A BLANK STATE



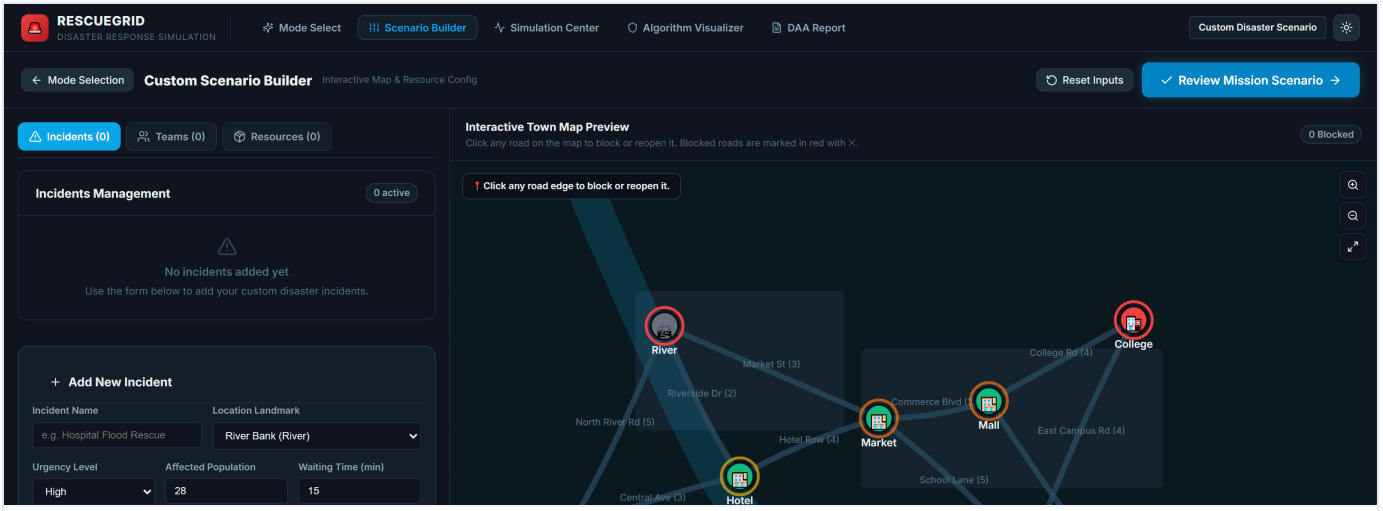
Functionality: Shows the clean user-input starting canvas with 0 pre-filled incidents, 0 teams, and 0 relief packages. Operators configure custom scenarios from a clean state.

Figure 3: Configured Custom Scenario — OPTION A USER INPUT



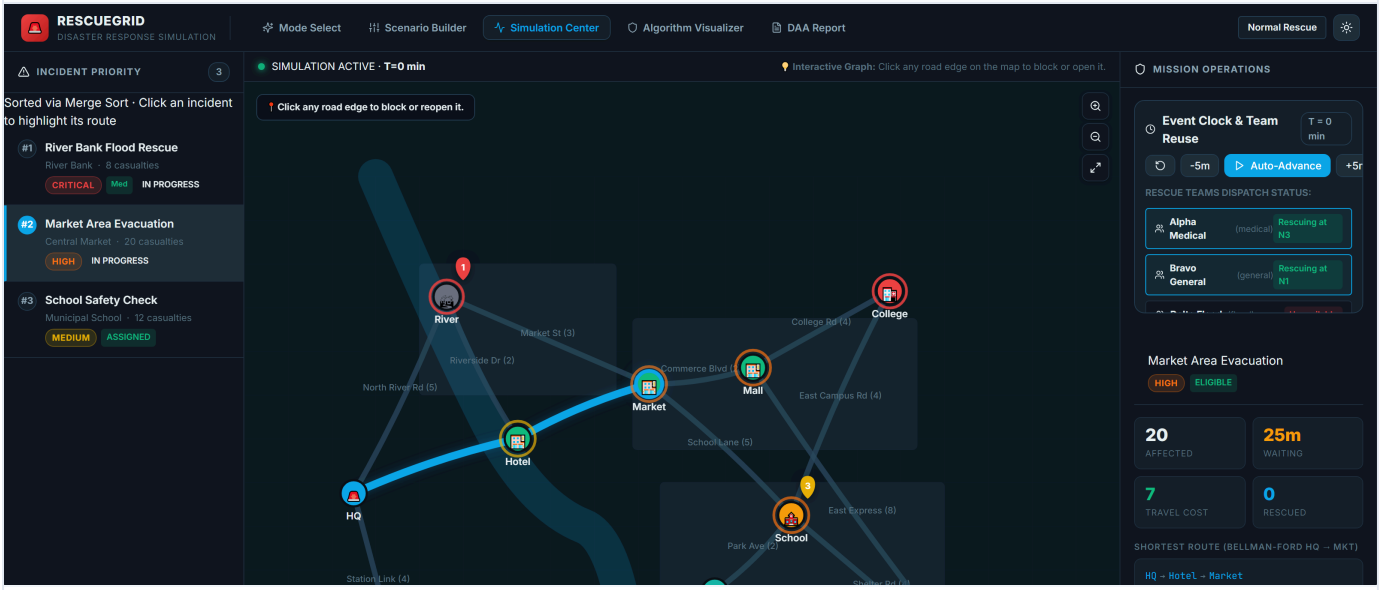
Functionality: Custom scenario populated with user-added incidents, paramedic teams, relief cargo, and active road blockages toggled directly by clicking road segments on the SVG map.

Figure 4: Pre-Flight Mission Verification & Parameter Review Modal — PRE-FLIGHT AUDIT



Functionality: Validates scenario integrity before algorithm execution. Computes total affected casualty demand, verifies team readiness, inspects road hazard blockages, and confirms vehicle payload constraints.

Figure 5: Operations Command Dashboard (Full Simulation Center) COMMAND CENTER



Functionality: The central operational hub. Integrates the left Merge-Sort triage queue, the interactive central SVG town map with active transit route highlights in cyan, and the right panel displaying Knapsack cargo and team dispatch.

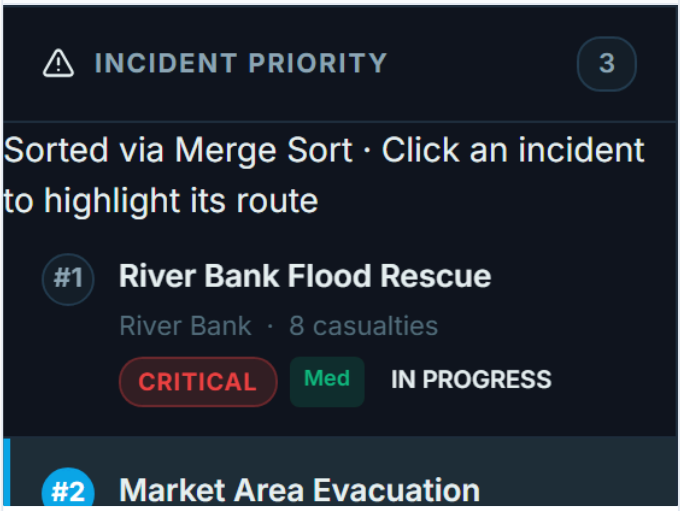
3. Application Screenshots (Core Subsystems Detail)

Figure 6: Town Road Network & Graph Detail SVG GRAPH



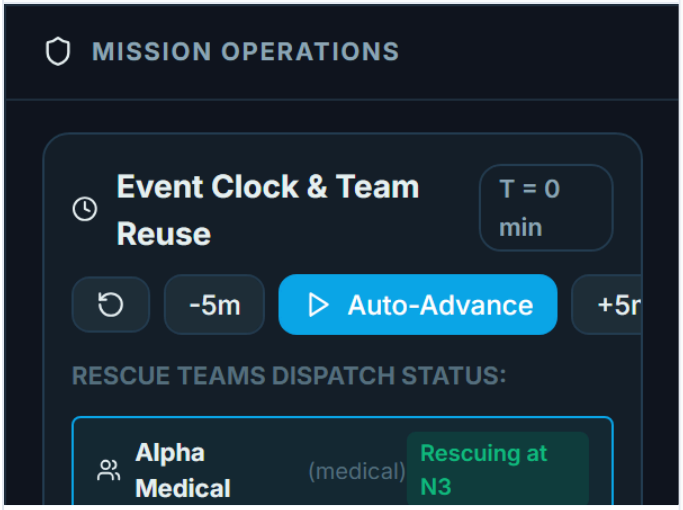
Functionality: High-detail graph view showing 13 municipal nodes, transit costs in minutes, blocked roads highlighted in red, and the active Bellman-Ford shortest route in neon cyan.

Figure 7: Incident Priority Queue MERGE SORT QUEUE



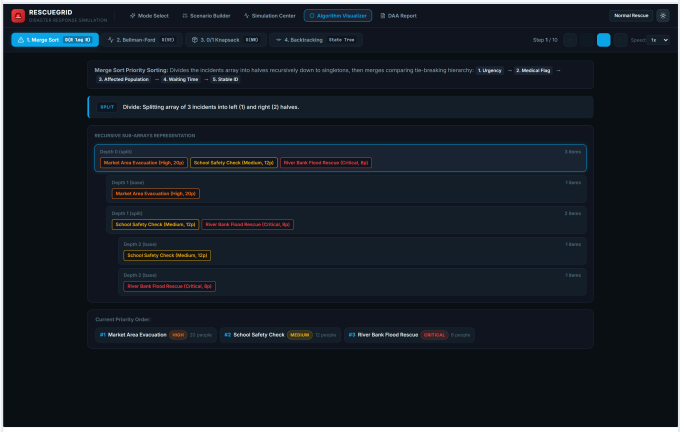
Functionality: Left sidebar displaying incidents ordered by the 5-tier Merge Sort comparator with rank badges (#1, #2), urgency chips, medical markers, and live fulfillment statuses.

Figure 8: Relief Cargo Packaging & Team Details 0/1 KNAPSACK PANEL



Functionality: Displays 0/1 Knapsack payload utilization (15/15 units, delivering 49 casualty benefit points), assigned team (Bravo General), and shortest path metrics.

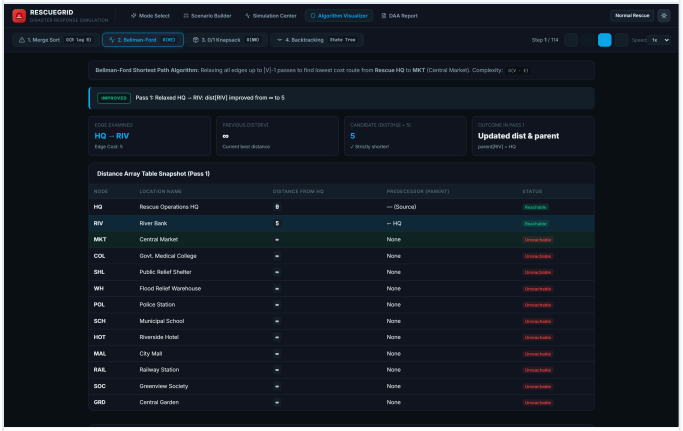
Figure 9: Merge Sort Visualizer DIVIDE & CONQUER



Functionality: Step-by-step trace showing recursive divide-and-conquer array halving, pairwise comparisons, and tie-breaking rationale for incident priority decisions.

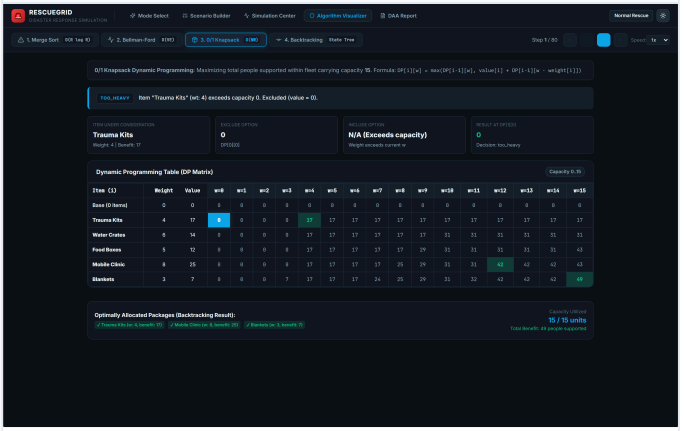
3. Application Screenshots (Algorithm Visualizers & Engineering Audit)

Figure 10: Bellman-Ford Edge Relaxation Visualizer DYNAMIC PROGRAMMING



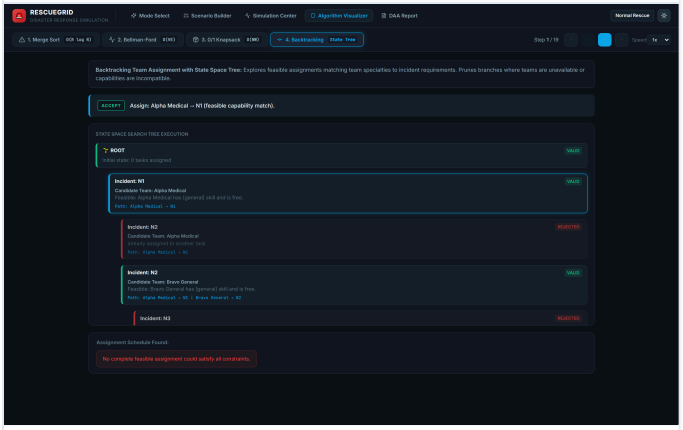
Functionality: Step-by-step edge relaxation inspector displaying current passes, candidate distance calculations, relaxed edges, and the live distance vector snapshot table.

Figure 11: 0/1 Knapsack DP Table Visualizer DP STATE MATRIX



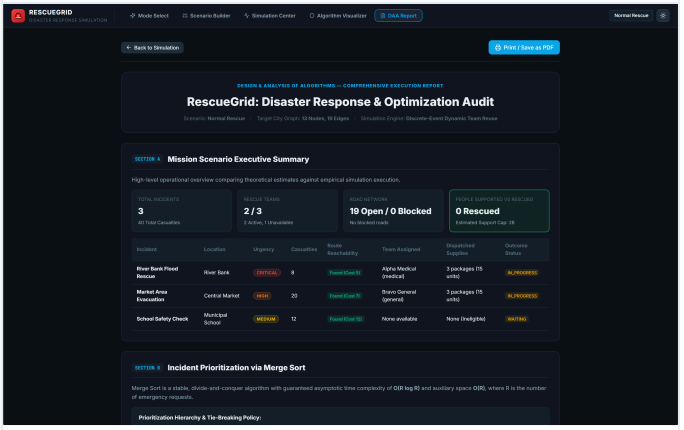
Functionality: Interactive (N+1)×(W+1) dynamic programming matrix rendering include vs exclude decisions for relief packages, complete with yellow backtracking path highlights.

Figure 12: Backtracking Team Assignment Visualizer STATE SPACE TREE



Functionality: Displays the combinatorial state space tree. Green nodes represent feasible skill assignments; red nodes indicate constraint violations pruned immediately.

Figure 13: DAA Execution Audit & Engineering Report PRINTABLE AUDIT



Functionality: Printable engineering report detailing operational KPIs, incident outcome tables (reachability, cargo dispatched, team assigned), road safety audits, and algorithm performance.

> **Verification & Test Suite Coverage:** All four algorithm implementations and discrete-event simulation transitions are verified by 28 automated tests running in Vitest (17 algorithm tests and 11 simulation tests, 100% passing).

4. Conclusion

RescueGrid demonstrates the direct application of fundamental Design and Analysis of Algorithms (DAA) concepts to life-critical humanitarian relief challenges. By formalizing real-world disaster coordination into well-defined mathematical problem instances:

- **Deterministic Incident Triage:** Multi-criterion Merge Sort ensures that urgency level, life-threatening medical conditions, and high casualty counts dictate response order without subjective triage errors.
- **Resilient Safe Routing:** Bellman-Ford guarantees optimal travel routes from Headquarters to disaster zones while systematically bypassing impassable hazards and bridge collapses.
- **Maximum Relief Impact:** 0/1 Knapsack Dynamic Programming achieves mathematically optimal vehicle payload utilization, outperforming greedy heuristics and preventing vehicle overloading.

• **Constraint-Satisfaction Dispatch:** Backtracking eliminates capability mismatches (e.g. dispatching general units to surgical emergencies) and double-booking through systematic pruning.

The system seamlessly integrates algorithm execution with an interactive SVG town map, comprehensive visual step-by-step traces, a clean custom scenario builder, and automated regression testing. RescueGrid stands as a robust, production-grade academic software project highlighting how algorithmic efficiency directly saves lives during municipal crises.